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ITAI 3377

9 April 2026

IIoT Network Analysis: Age of Information and Reliability Trade-offs

Age of Information (AoI) refers to how fresh or up to date data is when it reaches its destination. It measures the amount of time that has passed since the data was originally generated. This concept is important in Industrial Internet of Things (IIoT) systems because outdated data can lead to incorrect decisions. For example, in a smart factory, if a sensor sends temperature data but it arrives late, the system may assume everything is normal when the machine could already be overheating.

AoI-oriented traffic focuses on continuously receiving the most up to date information. For example, sensors that monitor temperature or pressure require constant updates to remain accurate. In contrast, deadline-oriented traffic focuses on ensuring that data arrives before a specific time. An example of this would be an emergency shutdown signal, which must arrive within a certain time frame to be effective. While AoI emphasizes data freshness, deadline-oriented traffic emphasizes timing constraints.

The dataset used in this analysis was loaded using pandas and explored to understand its structure. The first few rows, dataset information, and summary statistics were reviewed to identify patterns and characteristics of the data. Several visualizations were created to further

analyze relationships between variables. A scatter plot was used to observe relationships between selected features and Age of Information, a box plot was used to analyze the distribution of AoI values, and a heatmap was used to identify correlations between numerical variables.

```
import seaborn as sns
```

```
# Scatter plot
```

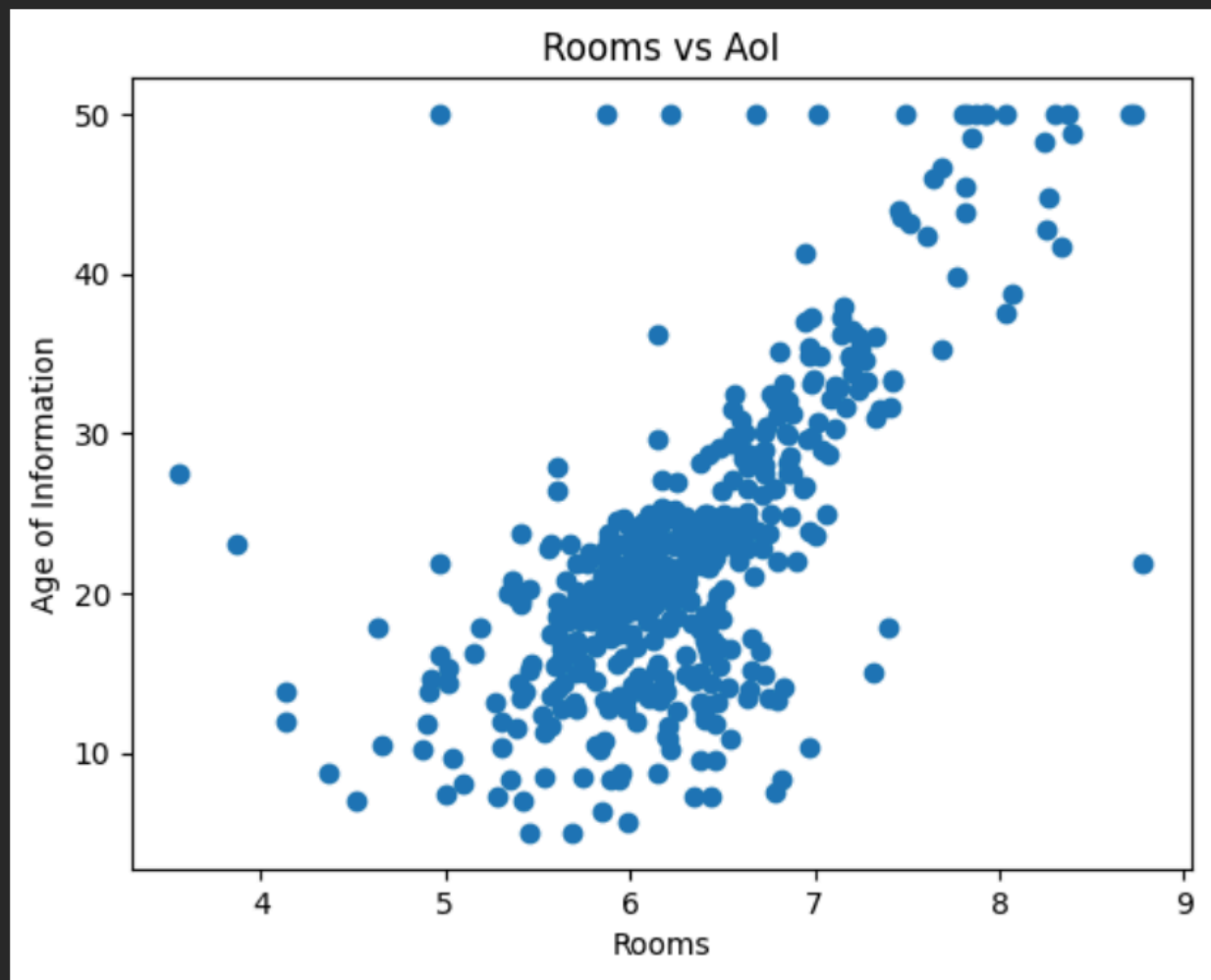
```
plt.scatter(df['rm'], df['age_of_information'])
```

```
plt.xlabel('Rooms')
```

```
plt.ylabel('Age of Information')
```

```
plt.title('Rooms vs AoI')
```

```
plt.show()
```

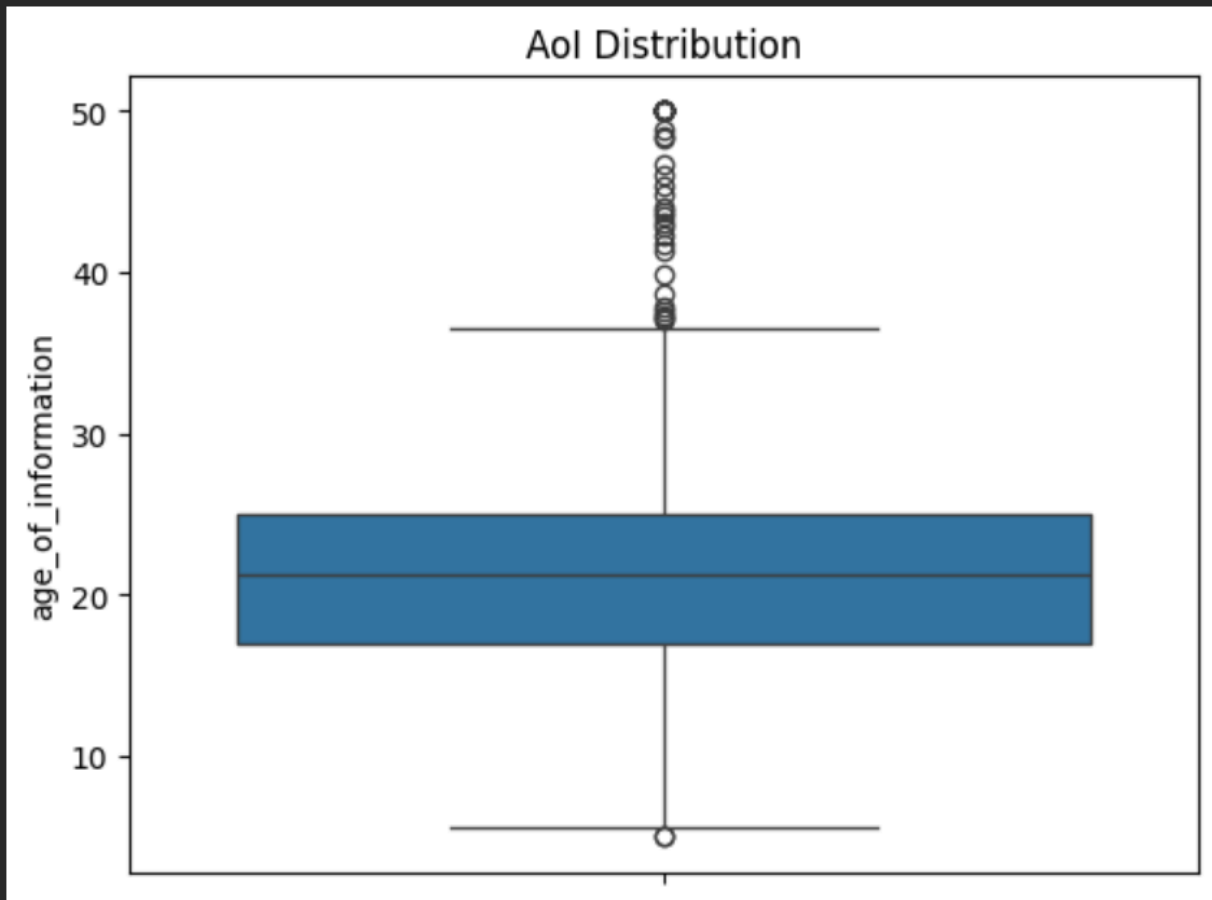


From these visualizations, several patterns were observed. First, some variables showed a relationship with Age of Information, suggesting that certain factors influence data freshness. Second, the heatmap indicated that some variables were more strongly correlated with AoI than others. Finally, the distribution of AoI values showed variability, indicating that different conditions affect how up to date the data is. These observations suggest that system conditions and input variables play an important role in both data freshness and reliability.

A Random Forest Regressor model was developed to predict the Age of Information. The dataset was divided into training and testing sets, and feature scaling was applied using StandardScaler. The model was evaluated using Mean Squared Error (MSE) and R squared (R^2). MSE measures the average difference between predicted and actual values, where lower values indicate better performance. R^2 measures how well the model explains the variation in the data, with values closer to 1 indicating a better fit. Feature importance analysis was also performed to

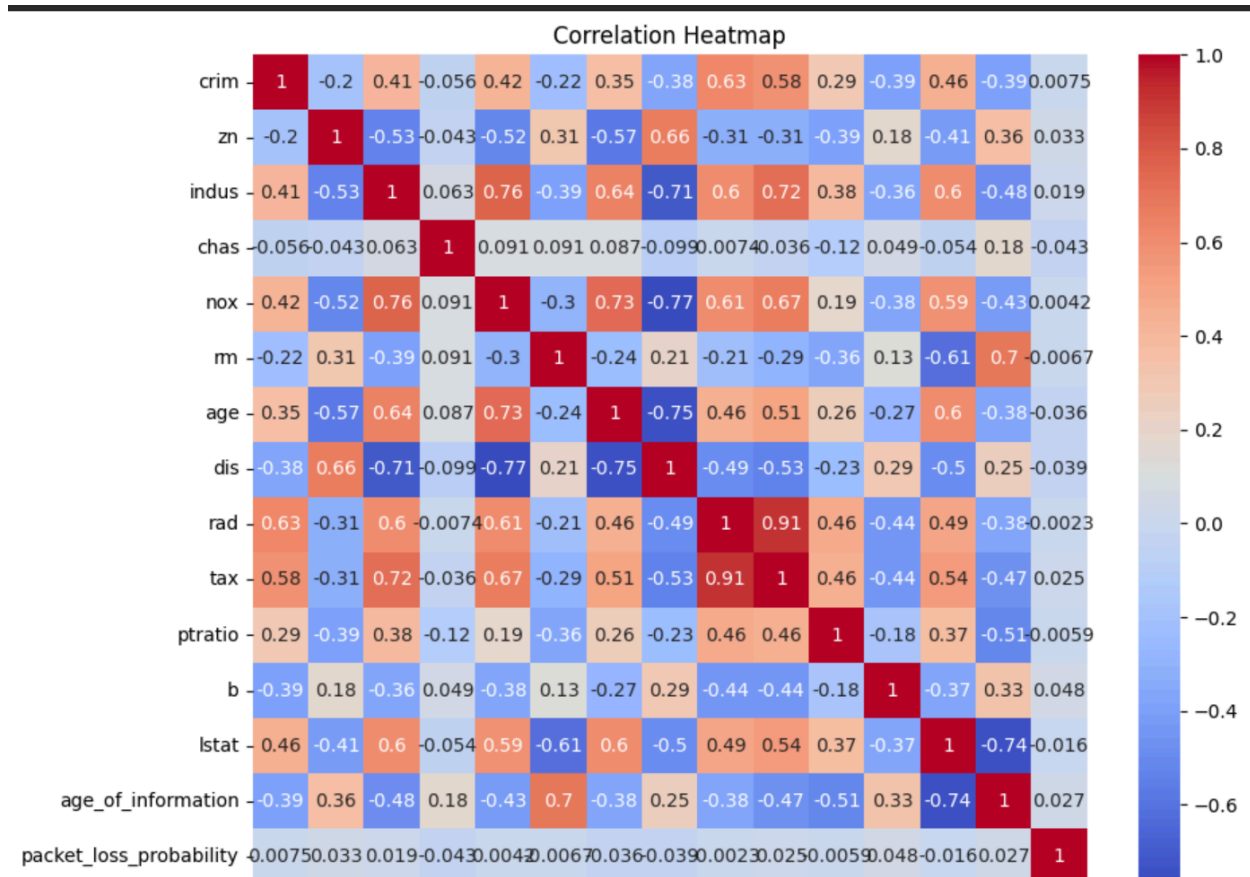
identify which variables had the most significant impact on AoI.

```
# Box plot  
sns.boxplot(y=df['age_of_information'])  
plt.title('AoI Distribution')  
plt.show()
```



The model was then used to make predictions on new data samples. The predicted values were consistent with the patterns observed earlier, indicating that the model was able to capture relationships between variables and the Age of Information. This demonstrates that machine

learning can be used to better understand and predict system performance in IIoT networks.



Several key factors influence the trade-off between AoI and reliability. System conditions and transmission-related variables directly affect how current the data is and whether it is delivered successfully. To optimize performance, increasing the frequency of data transmission can help reduce AoI and improve data freshness. Additionally, improving system reliability can reduce packet loss and ensure accurate data delivery.

These findings have important real world applications. For example, smart manufacturing systems rely on real-time monitoring to prevent equipment failure. Similarly, autonomous systems require timely and accurate data to operate safely. Understanding the relationship between AoI and reliability can help improve the performance of these systems.

In conclusion, this analysis highlights the importance of balancing data freshness and reliability in IIoT systems. By understanding how different variables influence AoI and packet loss, it is possible to improve system performance and support better decision-making in real-time environments.

Works Cited

Farag, H., et al. *On the Analysis of AoI-Reliability Tradeoff in Heterogeneous IIoT Networks*. arXiv, 2023.